

OFDM Term Project — Design Report

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Methods Implemented (M2 + M3 + M4 combined)

Pilot Pattern Rearrangement (TX modification) The default `dense_4sym` pattern places pilots only on symbols $\{0,3,6,9\}$, creating a 3-symbol gap of $130\ \mu\text{s}$. At $f_d = 3000\ \text{Hz}$, $J_0(2\pi \cdot 3000 \cdot 130 \times 10^{-6}) \approx 0$ — the channel is fully decorrelated between snapshots, making any interpolation useless. I changed to `comb_10sym_64`: 64 pilots distributed across *all* 10 symbols (even RBs: 3 pilots/RB at 4-SC spacing; odd RBs: 2 pilots/RB at 6-SC spacing; average 4.9 SC spacing, 248 data SCs). The reduced gap of $43\ \mu\text{s}$ gives $J_0(2\pi \cdot 3000 \cdot 43 \times 10^{-6}) \approx 0.84$ — strong correlation every symbol. The 64-pilot cap is respected exactly.

Residual STO Correction (M3 — Classen–Meyr phase-slope estimator). I implemented a frequency-domain differential phase estimator. A residual STO of δ samples imprints a linear phase ramp $e^{-j2\pi k\delta/N_{\text{FFT}}}$ across subcarriers. For each consecutive pilot pair with known frequency spacing Δk , $\delta = -\Delta\phi \cdot N_{\text{FFT}}/(2\pi \Delta k)$. Averaging over all valid pairs across all 10 symbols gives a robust estimate. This was chosen over CP self-correlation because the CP correlator has a flat-top ambiguity that makes it unreliable for small offsets. Unambiguous range is ± 128 samples, exceeding the ± 32 -sample spec.

Residual CFO Correction (M2 — van de Beek CP correlation). I used the **van de Beek CP self-correlation** estimator. A CFO Δf introduces a phase shift between the CP and its tail copy proportional to $2\pi \Delta f N_{\text{FFT}}/F_s$. Averaging over all 10 symbols gives range $\pm 15\ \text{kHz}$, covering $\pm 12\ \text{kHz}$. The method is unbiased under multipath because all channel taps (max 12 samples) fall well within the 288-sample CP.

Doppler-Aware Channel Estimation (M3 — scattered pilots + Wiener smoother). Two upgrades over LS + nearest-neighbour: (1) Cubic **spline** frequency interpolation handles the non-uniform 4/6 SC pilot spacing better than linear. (2) **Wiener time smoother** applied to all 10 LS pilot snapshots before interpolation: $\mathbf{W} = \mathbf{R}_{HH}(\mathbf{R}_{HH} + \sigma^2 \mathbf{I})^{-1}$, where $R_{HH}(i, j) = J_0(2\pi f_d |i - j| T_{\text{sym}})$ encodes Jakes temporal correlation. At $f_d = 0$ this averages all snapshots (noise $\downarrow \sqrt{10}$); at $f_d = 3000\ \text{Hz}$ it acts as a Jakes-shaped low-pass filter. $\sigma^2 = 5 \times 10^{-4}$ is tuned to the 30 dB grading point.

ICI-Aware Equalization (M4 — extended to Q=6 neighbours). At $f_d = 3000\ \text{Hz}$ the channel varies within one symbol ($T_u \approx 33\ \mu\text{s}$), causing inter-carrier interference. After initial ZF equalization, the channel slope ΔH per symbol is estimated from central differences of adjacent symbol H snapshots. ICI weights $w(\Delta k) = 1/(N(e^{-j2\pi \Delta k/N} - 1))$ are derived from the DFT of the linear time ramp. Hard QPSK decisions are used to subtract estimated ICI from Y before re-equalization, truncated to $Q = 6$ neighbours ($\approx 94\%$ of ICI energy).

M1 not used. M1's Schmid–Cox/Moose CFO estimator requires a preamble. Since this project operates on the data frame only (no preamble), M1's CFO mechanism is inapplicable. The Doppler tracking idea from M1 (per-symbol channel re-estimation from comb pilots) is adopted through the `comb_10sym_64` pilot pattern.

What Worked and What Didn't

The pilot rearrangement to `comb_10sym_64` was the single most impactful change, eliminating the Doppler dead zone caused by zero Jakes correlation in `dense_4sym`. The Wiener time smoother added +4% complexity for 13–19% BER reduction across all Doppler values. Several approaches were tried and reverted: *MMSE equalization* failed because LS estimation absorbs noise into the channel estimate ($\sigma^2 \rightarrow 0$), so the dominant error is interpolation bias, not noise amplification. *Wiener frequency interpolation* failed because the 711-bin DC/guard gap between the two active frequency halves makes the cross-half correlation matrix near-singular (coherence BW $\approx 400\ \text{kHz}$ vs. $9.2\ \text{MHz}$ gap). *Pilot-based CFO refinement* worsened performance by adding $\sim 200\ \text{Hz}$ Doppler phase noise, larger than the $< 50\ \text{Hz}$ van de Beek residual. *Decision-directed channel refinement* at all 312 SCs showed negligible gain because the dominant residual at $f_d = 3000\ \text{Hz}$ is non-linear Jakes variation within the symbol — a frequency-domain refinement does not address a time-domain problem.

Complexity — MATLAB Profiler (Total time: 0.091 s / frame)

Function	Total time (s)	% of frame
<code>sto_estimate_correct</code>	0.042	46%
<code>channel_estimate</code>	0.030	33%
<code>equalize</code>	0.010	11%
<code>cfo_estimate_correct</code>	0.002	2%
<code>cp_remove + ofdm_demod + qpsk_demap</code>	0.004	4%
<code>ofdm_rx</code> (total)	0.091	100%

Note: 0.035 s of the STO block is the `rng` call inside pilot regeneration (`local_pilot_values`), not algorithmic cost. The ICI cancellation loop (`equalize`: 0.010 s) is modest relative to channel estimation overhead.

Op-count estimate: Since `flops()` was removed in MATLAB R13, op-count was estimated by calibrating multiply-accumulate throughput on the test machine ($1.02 \times 10^9\ \text{MACs/s}$), then scaling by the algorithmic frame time (0.056 s = profiler total – `rng` overhead). **Estimated op-count: $\sim 5.7 \times 10^7\ \text{MACs/frame}$.**

BER Results at $f_d = 3000$ Hz (grading operating point)

f_d (Hz)	Tier-1 @ 30 dB	This receiver @ 30 dB	Improvement
1000	$\sim 1.54 \times 10^{-2}$	3.26×10^{-3}	$4.7\times$
2000	$\sim 4.50 \times 10^{-2}$	5.53×10^{-3}	$8.1\times$
3000	$\sim \mathbf{7.53 \times 10^{-2}}$	9.99×10^{-3}	$7.5\times$

Results obtained with N_TRIALS=200 (≥ 1400 errors per SNR point). BER plot submitted as **11han_ber.png**.